

FIRST PHASE

TEAM 6

LEXICAL ANALYZER

COMPILER PROJECT

WHAT IS A LEXICAL ANALYZER?



HOW DID WE BUILD OURS?

01



DEFINE

Tokens & Regular Expressions



02



MODEL

NFA → DFA



03



IMPLEMENT

Python Lexer

From theoretical concepts to a functional lexical analyzer

DEFINE THE LANGUAGE

TOKEN DEFINITION

45 TOKEN TYPES

7 TOKEN CLASSES

- KEYWORDS
- IDENTIFIERS
- CONSTANTS
- LITERALS
- OPERATORS
- PUNCTUATION
- SPECIAL CHARACTERS

REGULAR EXPRESSIONS

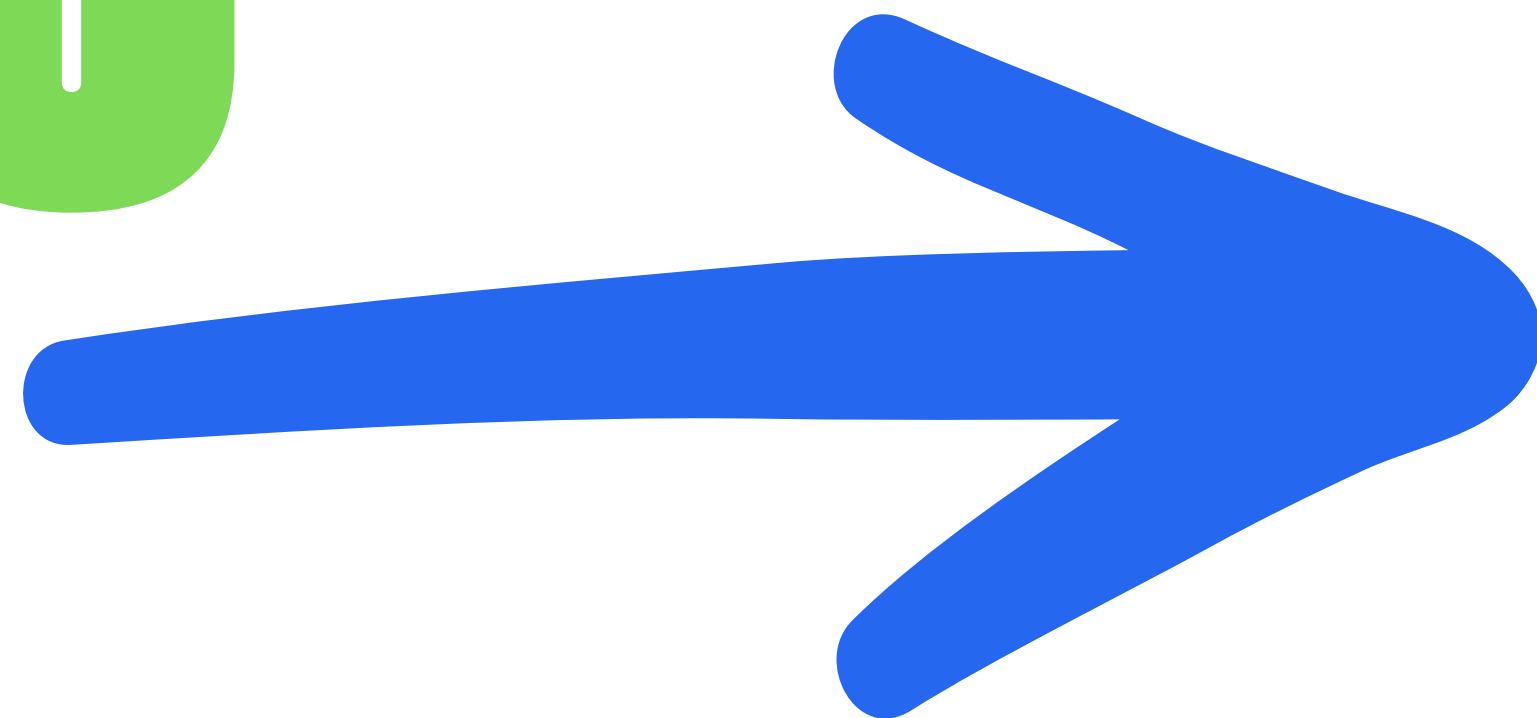
RULES THAT DEFINE HOW EACH TOKEN IS FORMED

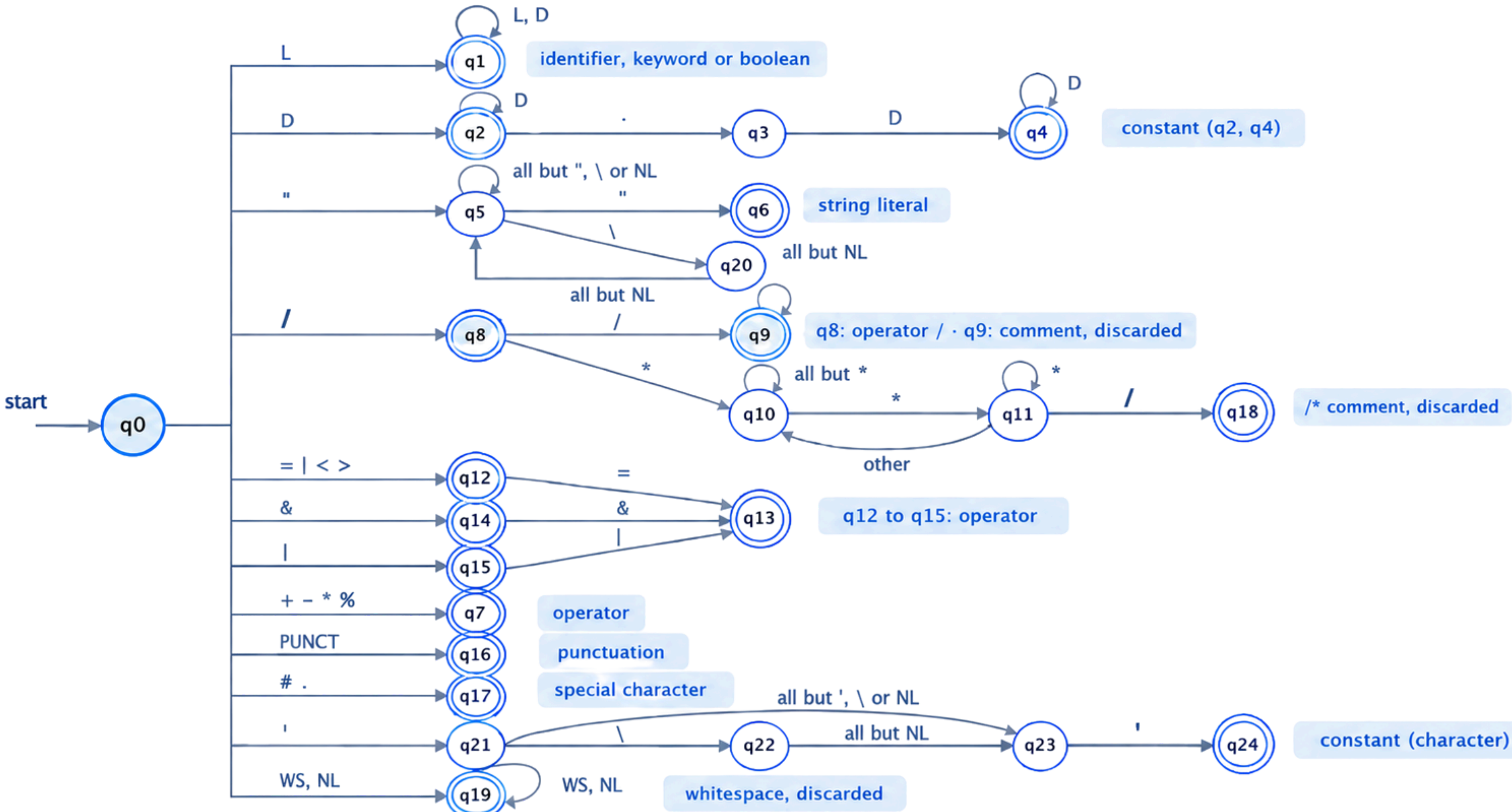
IDENTIFIER: STARTS WITH A LETTER OR `_` THEN: LETTERS, DIGITS OR `_`
 $(\text{LETTER} \mid _)(\text{LETTER} \mid \text{DIGIT} \mid _)^*$

INTEGER: ONE OR MORE DIGITS
 DIGIT^+

DECIMAL
 $\text{DIGITS} + \text{POINT} + \text{DIGITS}$
 $\text{DIGIT}^+ . \text{DIGIT}^+$

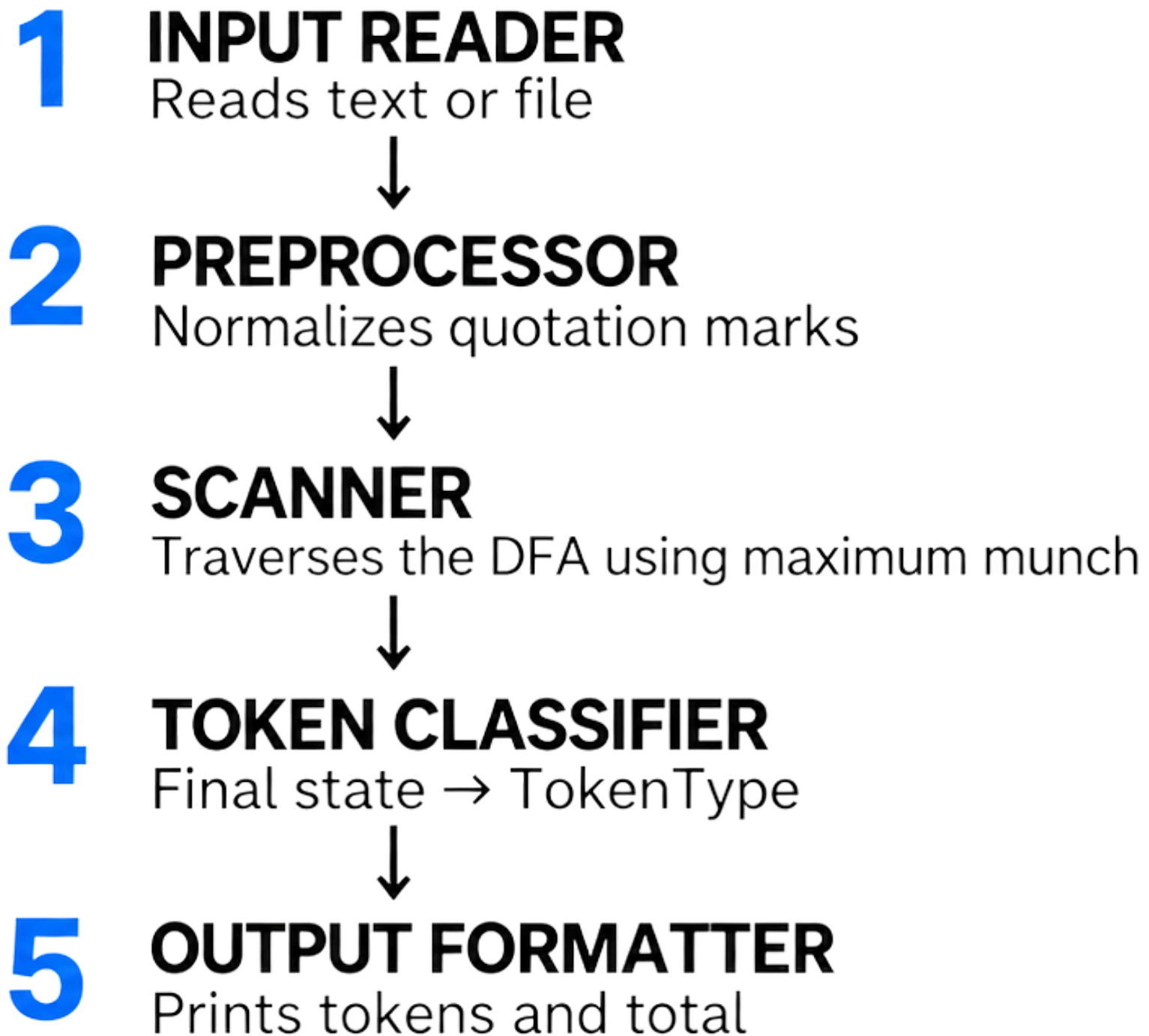
**FROM
REGEX TO
DFA**





IMPLEMENT THE LEXER

MAIN FLOW



SUPPORT MODULES

3 → ERROR HANDLER

- Called from Scanner
- Reports lexical errors
- Recovers when possible

4 → SYMBOL TABLE

- Used when the token is ID
- Stores identifiers
- Records occurrence lines

FROM TOKENS TO GRAMMAR

`int a = 10;`



INT
int

ID
a

ASSIGN
=

INT_CONST
10

SEMI
;

ORIGINAL CFG

Decl $\rightarrow$ **Type** **id** ;

Decl $\rightarrow$ **Type** **id** = **Value** ;

Type $\rightarrow$ int | float | char

Value $\rightarrow$ num | id



**LEFT
FACTORING**

LEFT FACTORING

Decl $\rightarrow$ **Type** **id** **Decl'**

Decl' $\rightarrow$;

Decl' $\rightarrow$ = **Value** ;

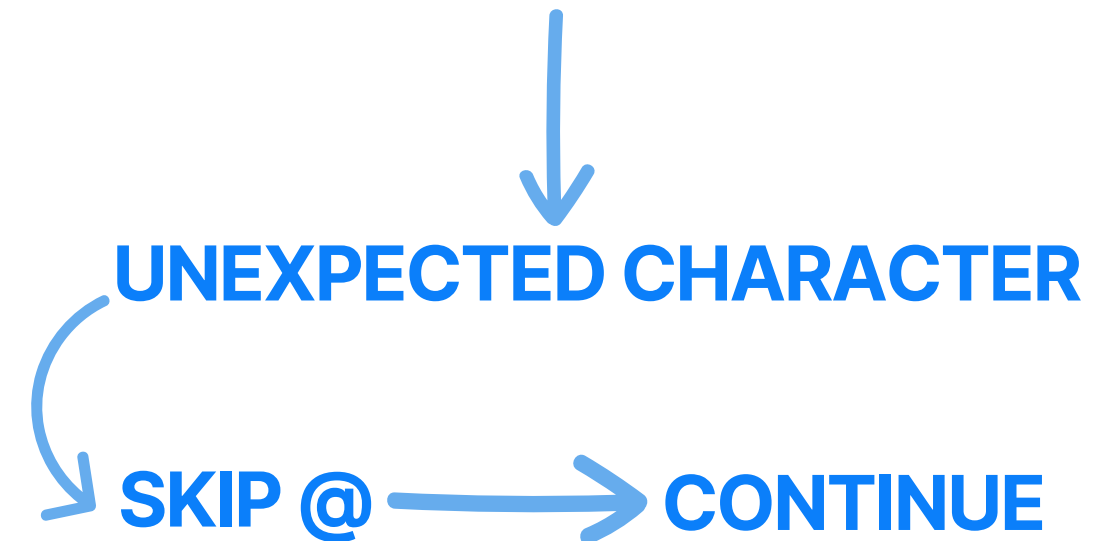
Type $\rightarrow$ int | float | char

Value $\rightarrow$ num | id

LEXICAL ERRORS & RECOVERY

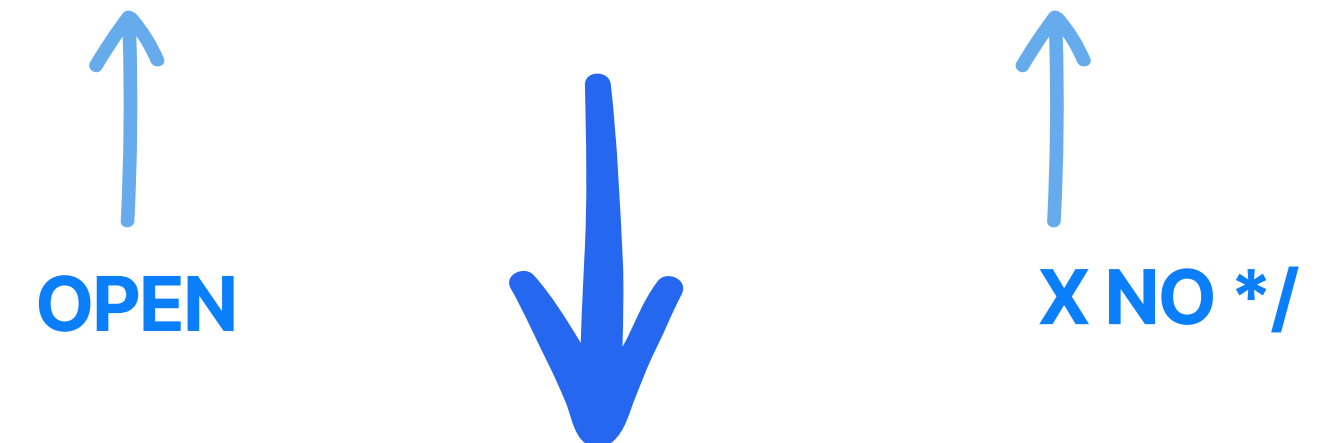
UNEXPECTED CHARACTER

INT @A = 10;



UNTERMINATED COMMENT

/* THIS IS A COMMENT.



UNTERMINATED STRING

"HELLO → UNTERMINATED STRING



UNTERMINATED COMMENT

STOP SCANNING

RUNNING TEST

```
C:\Users\ASUS\OneDrive\Desktop\7mo Semestre\Compilers\Entrega 1\unam.fi.compilers.g5.06>python main.py
--text "printf(\"This is an example\");"
identifier punctuation literal punctuation punctuation
Identifier: printf
Punctuation: (
Literal: "This is an example"
Punctuation: )
Punctuation: ;
Total of tokens: 5
```

Total of tokens: 5

FIRST PHASE RESULTS

45

TOKEN TYPES

25

DFA STATES

33

TESTS PASSED

**THANKS FOR
LISTENING**